$$\frac{x}{x-1} = \frac{1}{2} \text{ we get, } 2x = x - 1 \text{ implies } x = -1$$

$$\frac{x}{x-1} = 2 \text{ we get, } x = 2x - 2 \text{ implies } x = 2$$

Therefore, the roots are $x = -1,\ 2$.

Exercise 3.10

1. Solve the following quadratic equations by factorization method

   (i) $4x^2 - 7x - 2 = 0$ (ii) $3(p^2 - 6) = p(p + 5)$ (iii) $\sqrt{a(a-7)} = 3\sqrt{2}$

   (iv) $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ (v) $2x^2 - x + \frac{1}{8} = 0$

2. The number of volleyball games that must be scheduled in a league with n teams is given by $G(n) = \frac{n^2 - n}{2}$ where each team plays with every other team exactly once. A league schedules 15 games. How many teams are in the league?

Solving a Quadratic Equation by Completing the Square Method

In deriving the formula for the roots of a quadratic equation we used completing the squares method. The same technique can be applied in solving any given quadratic equation through the following steps.

Step 1: Write the quadratic equation in general form $ax^2 + bx + c = 0$.

Step 2: Divide both sides of the equation by the coefficient of x^2 if it is not 1.

Step 3: Shift the constant term to the right hand side.

Step 4: Add the square of one-half of the coefficient of x to both sides.

Step 5: Write the left hand side as a square and simplify the right hand side.

Step 6: Take the square root on both sides and solve for x.

Example 3.30 Solve $x^2 - 3x - 2 = 0$

Solution $x^2 - 3x - 2 = 0$

$$x^2 - 3x = 2 \qquad \text{(Shifting the Constant to RHS)}$$

$$x^2 - 3x + \left(\frac{3}{2}\right)^2 = 2 + \left(\frac{3}{2}\right)^2 \qquad \text{(Add } \left[\frac{1}{2}\left(\text{co-efficient of } x\right)\right]^2 \text{ to both sides)}$$

$$\left(x - \frac{3}{2}\right)^2 = \frac{17}{4} \qquad \text{(writing the LHS as complete square)}$$

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